

PROJECT 08

AI Mock Interview & Assessment Platform

Executive Abstract & Feature Priority Tiers

Version	1.0
Date	September 2026

P0 Core Conversational Interview

Mandatory baseline. The fundamental real-time AI interview experience.

- **Resume Intelligence.** PDF parsing, extracting candidate skills, projects, and verifiable claims to ground the interview context.
- **Real-Time AI Interviewer.** Role, resume, and JD-based technical and HR/behavioral questioning.
- **Stateful Interview Agent.** Autonomous agent loop (Observe → Reason → Decide → Act → Verify) managing conversation state and context.
- **Dynamic Follow-ups & Adaptive Difficulty.** Real-time probing of candidate answers, challenging weak assertions, and scaling question difficulty.
- **Evaluation & Scoring.** Transparent rubric scoring, strengths/weaknesses identification, and full session transcript history.

P1 Assessment Platform & In-Interview Coding

Formal examination infrastructure and agent-invoked live coding.

- **Interactive Coding Tool.** 1-on-1 coding tool dynamically invoked by the AI agent mid-interview; candidate writes code, Judge0 executes against test cases, AI evaluates logic, edge cases, and Big-O complexity.
- **Standalone Coding Assessment.** Formal scheduled exams configured by administrators with custom problem pools, languages, test durations, and auto-scoring.
- **Secure Exam Environment (SEB).** Restricted/locked examination environment with fullscreen enforcement, tab-switch tracking, and unauthorized access limits.
- **Judge0 Sandbox Execution Engine.** Secure sandbox executing multi-language code (Python, Java, C++, JS) with 2.0s CPU timeout and 128MB memory limit.

P2 Advanced Assessment Modes & Intelligence

Specialized technical evaluations, voice interfaces, and long-term analytics.

- **System Design Assessment.** AI-led architecture round evaluating APIs, database trade-offs, scalability, caching, queues, and failure recovery.
- **Written / Notepad Round.** Free-form workspace for pseudocode, algorithm explanations, and SQL queries without execution.
- **Voice Interview.** Real-time speech-to-text and text-to-speech conversational voice agent.
- **Company-Specific Simulation.** Company-specific interview styles, cultural questions, and role-specific hiring rubrics (e.g., Amazon LP, Google DSA).
- **Competency Intelligence.** Granular sub-skill scoring (e.g., Java: 84%, SQL: 68%, Problem Solving: 81%, System Design: 52%).
- **Personalized Practice & Progress Tracking.** Automated 30-day remediation plans and longitudinal tracking of score improvements.

P3 Institutional & College Platform

Administrative deployment, cohort management, and reporting.

- **College Assessment Management.** Drive creation, target role configuration, interview conditions setup, and question pool governance.
- **Student Cohort Allocation.** Large-scale automated batching of students across tracks (e.g., 100 students → Coding Exam vs AI Interview vs HR).
- **Faculty & Admin Analytics Dashboard.** Aggregate placement readiness metrics, department-level comparisons, candidate score distribution, and audit logs.

Tier	Focus Area	Key Deliverable
P0	Core Product	Real-time adaptive AI mock interview + Resume intelligence
P1	Assessments	Standalone SEB coding exams + In-interview interactive coding tool
P2	Advanced AI	System design, voice interviews, and multi-session skill tracking
P3	Institutional	College administration, student cohort allocation, faculty analytics